

Shubham

Software Engineer | AI Systems | Secure Backends

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github.com/shubhamdahiya210902-tech
Open to working-student and internship roles in Germany | MSc expected Mar 2027



PROFILE SNAPSHOT

MSc Applied Research in Computer Science student and Research Assistant building Python/FastAPI services, LLM-assisted review workflows, PostgreSQL-backed data systems, and secure API integrations for applied research teams.
Python | FastAPI | PostgreSQL | AWS | LangGraph | LangChain Core | LiteLLM | MISP APIs | PyTorch | scikit-learn

EXPERIENCE

Research Assistant - Hof University of Applied Sciences

Oct 2024 - Present | Applied research software, data workflows, secure integrations
- Built Python REST services, SQL-backed workflows, and automation pipelines for 5+ applied research projects.
- Worked with PostgreSQL, MariaDB-Galera, AWS-backed infrastructure, documentation, validation checks, and stakeholder summaries.
- Built Thunderbird and Outlook phishing-reporting workflow connected to MISP through secure REST APIs, reducing manual handling by about 40%.

Independent Vulnerability Research - Responsible disclosure

Security research | Reports covering root cause, impact, reproduction, remediation, and verification
- Received three Government of India Letters of Acknowledgement for responsible vulnerability disclosure.

SELECTED PROJECTS

LangGraph Review Workflow

Local workflow for requirement extraction, evidence mapping, coverage scoring, groundedness checks, and human approval before generated application material is used.

FastAPI Geospatial Data Service

AWS-hosted FastAPI service serving 10GB+ historical flight and weather data with PostgreSQL-backed filtering, validation, and query-efficiency improvements.

Natural-Language-to-Code Research

Transformer-based prototype for converting natural-language queries into executable Python and SQL; SmartCom 2024 publication.

Traffic Forecasting Pipeline

Compared LSTM, ARIMA, and regression baselines on 1.2M+ GPS trajectory points for urban mobility forecasting.

EDUCATION

MSc Applied Research in Computer Science - Hof University of Applied Sciences

Expected Mar 2027 | Generative AI, multi-agent systems, intelligent automation, machine learning, cloud, applied research
Transcript supports Data Science wording through Data Engineering and Analysis plus Data Mining and Machine Learning modules.

ROLE FIT

AI / LLM Working Student	LangGraph review workflow, scoring rubrics, groundedness checks, SmartCom 2024 NL2Code research.
Python Backend Intern	FastAPI, PostgreSQL, REST APIs, AWS-hosted services, validation checks, documentation.
Security Automation Role	MISP integrations, token-scoped APIs, phishing reporting, Cisco CyberOps, responsible disclosure.

LANGUAGES AND LOGISTICS

English C2 | German A2 improving | Hindi native | Munich-based | Germany-wide remote or hybrid for strong fits | Student work authorization in Germany
Portfolio: shubhamdahiya210902-tech.github.io/personal-portfolio/